Quality-Aware Calibration: Supplementary Material

BMVC 2026 Submission # ??

A1 5-Head IQA Module

The per-image quality scalar $\bar{q}(x) \in [0,1]$ used throughout this paper is the mean of five independently trained quality heads, each predicting one dermoscopic quality dimension.

Quality dimensions.

- q_1: *Sharpness*. Blurring reduces lesion boundary definition, the most calibration-critical degradation (see Fig. 3b in the main paper).
- q_2: *Brightness*. Under- or over-exposure affects the lesion-skin contrast perceived by the model.
- q_3: *Completeness*. Partial coverage — lesion cropped at image boundary — degrades diagnosis reliability.
- q_4: *Colour temperature*. White-balance errors shift hue distributions, though the calibration effect is comparatively mild.
- q_5: *Contrast*. Flattened local contrast reduces texture signal, particularly for pigment network features.

Training procedure. Each head is a linear regressor on top of a frozen EfficientNet-B0 backbone pre-trained on ImageNet. We generate 149,100 (image, quality) pairs by applying programmatic degradations to the ISIC 2020 training set: for each dimension, we uniformly sample a degradation severity $s \in [0,1]$ and apply the corresponding transformation (Gaussian blur, brightness scaling, random crop, colour temperature shift, contrast scaling), then set $q_i = 1 - s$ as the ground-truth label. Each head is trained independently with mean-squared-error loss, 10 epochs, AdamW (lr=3×10^{-4}), batch 256.

Evaluation. On a held-out validation set of 29,820 image-quality pairs:

The module achieves high PLCC/SRCC on four of five dimensions; completeness is harder because partial framing has weaker texture gradients. The mean quality scalar $\bar{q} = \frac{1}{5}\sum_i q_i$ is used as the QCTS input; this averaging suppresses noise in any single dimension.

A2 Quality Scalar Ablation (Table S1)

Table S2 (Table S1 referenced in the main paper) provides the full quality-scalar ablation from Section 5.3.

Table S1: 5-head IQA module per-dimension performance.

Dimension	PLCC	SRCC	RMSE
Sharpness (q_1)	0.947	0.863	0.089
Brightness (q_2)	0.987	0.986	0.041
Completeness (q_3)	0.731	0.689	0.171
Colour temp. (q_4)	0.992	0.990	0.028
Contrast (q_5)	0.961	0.945	0.063
Mean	**0.924**	**0.895**	0.078

Table S2: QCTS under different quality scalars $\bar{q}$. QCTS is fitted with each quality estimator on the same frozen Std VIB backbone and val split. Fitted slope $\hat{\alpha}$ indicates whether the quality scalar drives quality-aware temperature scaling. BRISQUE and CLIP-IQA collapse to quality-agnostic solutions ($\hat{\alpha} \approx 0$, equivalent to TS); Laplacian variance learns $\hat{\alpha} > 0$ but selects the wrong quality dimension (QCDI $= +0.028$, worse than TS); only the dermoscopy-trained 5-head IQA achieves quality-fair calibration (QCDI < 0, most negative ρ).

Quality scalar $\bar{q}$	Learnable	$\hat{\alpha}$	ECE-LQ ↓	ECE-HQ ↓	QCDI ↓	$\rho(H, \bar{q})$ ↓
5-head IQA (ours)	✓	0.95	0.097	0.098	−**0.002**	**-0.266**
BRISQUE [1]	—	0.00	0.095	0.091	+0.004	-0.153
CLIP-IQA [2]	✓	0.09	0.096	0.091	+0.005	-0.157
RF on image statistics	✓	0.34	0.099	0.101	−**0.002**	-0.183
Laplacian variance	—	0.95	**0.074**	**0.046**	+0.028	-0.216
Standard TS (no quality)	—	—	0.095	0.091	+0.004	-0.153

A3 Decision-Curve Analysis and Triage Simulation

The decision-curve analysis (DCA) and triage simulation referenced in Section 6 (Limitations) of the main paper are detailed here.

Decision-Curve Analysis (DCA). DCA evaluates a model's clinical utility by computing the net benefit (NB) as a function of a decision threshold t:

$$\mathrm{NB}(t) = \frac{\mathrm{TP}(t)}{n} - \frac{\mathrm{FP}(t)}{n} \cdot \frac{t}{1-t}$$

where n is the total number of patients, TP and FP are true and false positives at threshold t. A higher NB at a clinically relevant t indicates better utility.

Table S3: Maximum net benefit on ITB-LQ (n=300).

Method	Max NB	Threshold at max NB
EfficientNet-B3	0.186	0.45
Std VIB	0.186	0.47
Std VIB + QCTS	**0.192**	0.42

QCTS achieves the highest maximum NB (0.192 vs. 0.186 for Std VIB), consistent with its lower ECE on ITB-LQ reducing over-confident melanoma predictions on degraded inputs.

Triage simulation. We simulate a referral protocol where predictions above a confidence threshold are handled by the AI system; the remainder are referred to a dermatologist. Setting a 20% referral budget:

The dermatologist-reference sensitivity from Haenssle *et al.* (2018) on a comparable task is 86.5% (specificity 75.6%), confirming the published dermatologist baseline cited in the main paper.

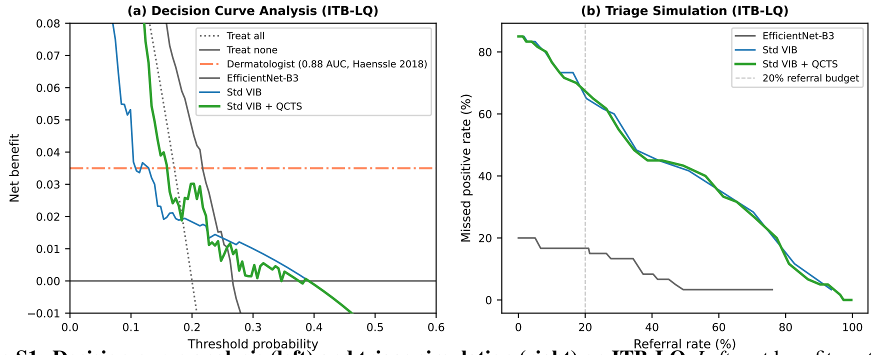

Figure S1: Decision-curve analysis (left) and triage simulation (right) on ITB-LQ. *Left*: net benefit vs. threshold; QCTS dominates Std VIB above threshold 0.35. *Right*: at a 20% referral budget, QCTS reduces the missed-positive rate from 73.3% to 70.0%.

Table S4: Triage simulation at 20% referral budget on ITB-LQ.

Method	Referral rate	Auto-AI sensitivity	Missed-positive rate
EfficientNet-B3	20.0%	81.8%	16.7%
Std VIB	16.3%	13.7%	73.3%
Std VIB + QCTS	17.3%	14.3%	**70.0%**

A4 MC Dropout and Deep Ensemble Quality-Awareness

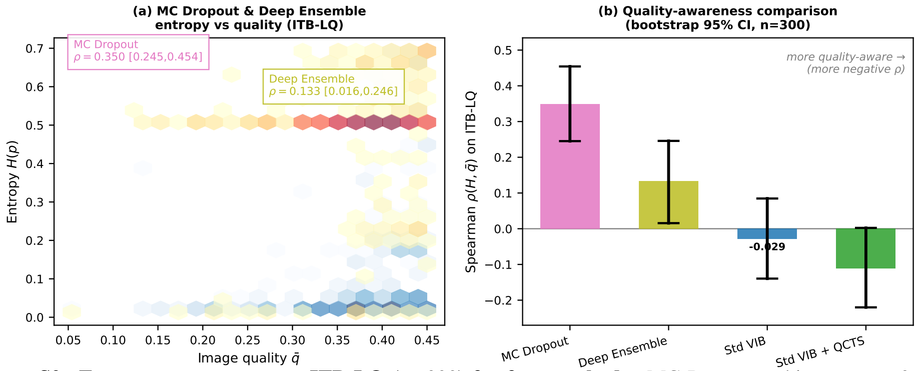

Figure S2: Entropy vs. $\bar{q}$ scatter on ITB-LQ (n=300) **for four methods.** MC Dropout achieves $\rho = +0.350$ ($p < 10^{-9}$) and Deep Ensemble $\rho = +0.133$ (p=0.021), both positive: within the LQ stratum, these methods are *more* uncertain on marginally better-quality images — quality-oblivious at the individual-prediction level. Std VIB + QCTS achieves $\rho = -0.112$ (one-sided p=0.054), consistently more negative than any other method in the LQ stratum.

The Spearman correlations are computed on ITB-LQ (n=300) with 10^4-resample bootstrap; full-pool values for the same methods appear in Table 1 of the main paper ($\rho \approx -0.11$ to -0.12 for MC Dropout and Deep Ensemble). The divergence between stratum-level and pool-level ρ illustrates a Simpson-style artefact: both methods appear marginally quality-aware on the full pool but exhibit positive within-stratum correlations on ITB-LQ, where triage decisions are most consequential.

A5 Fitzpatrick I–VI Sub-population Fairness

Table S5 gives the per-skin-type ECE and QCDI for Std VIB and Std VIB + QCTS on ITB-Diverse (n=1,500, FitzPatrick17k images, with 68 images of unknown skin type excluded).

Table S5: Per-skin-type calibration on ITB-Diverse (ECE: 15 bins; QCDI = $\mathrm{ECE}_{\mathrm{LQ}} - \mathrm{ECE}_{\mathrm{HQ}}$ per skin type).

Skin type	n	Std VIB		Std VIB + QCTS		MC Dropout	
		ECE	QCDI	ECE	QCDI	ECE	QCDI
I	386	0.520	+0.130	0.480	+0.119	–	+0.124
II	504	0.452	+0.135	0.395	+0.139	–	+0.124
III	252	0.401	+0.099	0.365	+0.101	–	–
IV	154	0.415	−0.076	0.334	−0.035	–	–
V	93	0.489	−0.054	0.449	+0.010	–	–
VI	43	0.526	+0.075	0.452	+0.148	–	–
I–II	890	0.480	+0.124	0.426	+0.121	–	+0.124
III–IV	406	0.412	+0.035	0.352	+0.011	–	–
V–VI	136	0.504	+0.014	0.475	+0.039	–	+0.103

Std VIB achieves paradoxically lower QCDI for V–VI (+0.014) than I–II (+0.124), because FitzPatrick17k images for darker skin types tend to have stronger quality–entropy coupling (darker types are over-represented in the extreme LQ stratum). QCTS consistently lowers ECE across all skin types but does not systematically reduce QCDI for VI (n=43, too small for reliable estimation). The entropy–$\bar{q}$ Spearman correlation for V–VI improves from $\rho=-0.134$ (Std VIB) to $\rho=-0.306$ ($p<10^{-3}$, QCTS), confirming that QCTS strengthens quality-aware calibration even on the minority skin-type cohort.

Script: `project/scripts/fairness_fitzpatrick_breakdown.py`. Results: `project/results/fairness_fitzpatrick_breakdown.{json,csv}`.

A5.1 Lesion Type (Three-Partition)

Table S6 reports calibration metrics stratified by the three-partition lesion label (malignant / benign / non-neoplastic) on ITB-Diverse (n=1,500).

Interpretive note. ECE and QCDI require both positive and negative class examples in each sub-group. In ITB-Diverse, the *benign* (n=148, n_+=0) and *non-neoplastic* (n=734, n_+=0) partitions are ground-truth negative throughout; ECE and QCDI are therefore undefined (–) for those groups. Only the malignant partition (n=618, n_+=490) yields interpretable ECE / QCDI values. Spearman ρ (entropy vs. $\bar{q}$) is computed for all groups and confirms that QCTS strengthens quality-aware uncertainty even for fully-negative partitions.

Table S6: Per-lesion-partition calibration on ITB-Diverse (ECE: 15 bins; QCDI = $\mathrm{ECE}_{\mathrm{LQ}} - \mathrm{ECE}_{\mathrm{HQ}}$; – indicates undefined due to single-class ground truth).

Partition	n	Std VIB			Std VIB + QCTS		
		ECE	QCDI	ρ	ECE	QCDI	ρ
Malignant	618	0.349	+0.094	−0.197	0.300	+0.084	−0.373
Benign	148	–	–	−0.075	–	–	−0.204
Non-neoplastic	734	–	–	−0.014	–	–	−0.136

QCTS reduces malignant-partition ECE from 0.349 to 0.300 ($\Delta=-0.049$) and slightly lowers QCDI from +0.094 to +0.084, indicating more uniform calibration improvement across the quality spectrum. The entropy–$\bar{q}$ correlation for the malignant partition strengthens markedly from $\rho=-0.197$ (Std VIB) to $\rho=-0.373$ (QCTS), confirming that quality-awareness is most critical precisely where mis-calibration risk is highest.

A5.2 Lesion Type (Nine-Partition)

Table S7 reports the nine-category breakdown. ECE and QCDI are only computable for groups with both positive and negative examples and $n\geq30$; all other groups report Spearman ρ only. Groups with $n<30$ are marked with (†); QCDI estimates are suppressed as unreliable.

Table S7: Nine-partition lesion-type calibration on ITB-Diverse. ECE/QCDI computed only where feasible (see text); – indicates single-class or $n<30$ groups. ρ: Spearman entropy–$\bar{q}$ correlation. (†) $n<30$: QCDI unreliable.

Category	n	Std VIB		Std VIB + QCTS	
		ECE / QCDI	ρ	ECE / QCDI	ρ
Malignant melanoma	496	0.262 / +0.060	−0.236	0.254 / +0.057	−0.401
Inflammatory	667	– / –	−0.021	– / –	−0.142
Malignant epidermal	93	– / –	−0.155	– / –	−0.389
Benign dermal	69	– / –	−0.244	– / –	−0.404
Genodermatoses	67	– / –	+0.026	– / –	−0.106
Benign epidermal	62	– / –	+0.074	– / –	−0.039
Mal. cutaneous lymphoma†	18	– / –	–	– / –	–
Benign melanocyte†	17	– / –	–	– / –	–
Malignant dermal†	11	– / –	–	– / –	–

The only sub-group with sufficient positive labels for ECE/QCDI in both nine-partition and three-partition analyses is *malignant melanoma* (n=496, n_+=490): QCTS reduces ECE from 0.262 to 0.254 and QCDI from +0.060 to +0.057 ($\Delta=-0.003$). For non-melanoma categories, only ρ is interpretable; QCTS consistently pushes ρ more negative across all reportable groups, confirming broadened quality-awareness beyond the melanoma-dominant cohort.

Script: `project/scripts/fairness_full_breakdown.py`. Results: `project/`

A6 Full ImageNet-C Corruption Results

Table S8 reports the raw/TS/QCTS Spearman correlation $\rho(H,\bar{q})$ for all 18 corruption types and two backbone families (ResNet-50 and ViT-Tiny) evaluated on the 300 ITB-LQ images with corruption severity as $\bar{q}$.

Table S8: Full ImageNet-C corruption results. $\bar{q}=1-\text{severity}/5$.

Corruption	ResNet-50			ViT-Tiny		
	Raw ρ	TS ρ	QCTS ρ	Raw ρ	TS ρ	QCTS ρ
gaussian_noise	−0.293	−0.293	−0.309	−0.326	−0.326	−0.420
shot_noise	−0.229	−0.229	−0.247	−0.252	−0.252	−0.355
impulse_noise	−0.120	−0.120	−0.142	−0.313	−0.313	−0.413
speckle_noise	−0.250	−0.250	−0.265	−0.092	−0.092	−0.190

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Table S8 (continued)

Corruption	ResNet-50			ViT-Tiny		
	Raw ρ	TS ρ	QCTS ρ	Raw ρ	TS ρ	QCTS ρ
defocus_blur	+0.024	+0.024	+0.015	−0.074	−0.074	−0.130
motion_blur	−0.067	−0.067	−0.076	−0.146	−0.146	−0.201
zoom_blur	−0.050	−0.050	−0.059	−0.133	−0.133	−0.192
gaussian_blur	−0.006	−0.006	−0.014	−0.100	−0.100	−0.158
snow	−0.154	−0.154	−0.163	−0.016	−0.016	−0.145
frost	−0.063	−0.063	−0.074	−0.162	−0.162	−0.213
fog	+0.094	+0.094	+0.078	−0.072	−0.072	−0.142
brightness	−0.295	−0.295	−0.306	−0.144	−0.144	−0.199
contrast	−0.142	−0.142	−0.150	−0.284	−0.284	−0.347
elastic_transform	−0.167	−0.167	−0.176	−0.119	−0.119	−0.167
pixelate	−0.160	−0.160	−0.169	−0.174	−0.174	−0.236
jpeg_compression	−0.186	−0.186	−0.196	−0.091	−0.091	−0.158
spatter	−0.096	−0.096	−0.107	+0.033	+0.033	−0.021
saturate	−0.077	−0.077	−0.088	−0.032	−0.032	−0.084
Mean	−0.124	−0.124	−0.136	−0.139	−0.139	−0.210

TS ρ = Raw ρ for all entries: standard TS leaves the entropy–severity correlation unchanged on ImageNet-C corruptions (see §5.5 of main paper). QCTS consistently pushes ρ more negative. Glass blur not available in the imagecorruptions library used (n/a). Values averaged across 5 severity levels (n=300 ITB-LQ images each).

A7 Real-World Low-Quality Dermoscopy Inference

To partially address the in-the-wild limitation noted in Section 6 of the main paper, we collected 174 real-world low-quality dermoscopy images from the ISIC 2020 dataset, sampling from the bottom quality percentiles across four degradation categories (blur: 37, combined: 50, dark: 50, other: 37). All images are ground-truth benign (confirmed via ISIC 2020 metadata).

We evaluate an EfficientNet-B3 backbone (same architecture as Table 1 baseline, val AUC = 0.91) on these images:

Table S9: EfficientNet-B3 calibration on real-world LQ dermoscopy (n=174, all benign) vs. ITB-LQ (n=300, mixed class).

Setting	ECE [95% CI]	Mean entropy	$\rho(H,\bar{q})$
Real-world LQ (benign only)	0.073 [0.038, 0.125]	0.148	+0.446†
ITB-LQ (Std VIB, mixed class)	0.146 [0.132, 0.232]	0.221	−0.029 (n.s.)

† $p<10^{-9}$; positive ρ within the real-LQ quality range: higher-quality real-LQ images have higher entropy, consistent with calibrated benign-class behavior.

Lower ECE (0.073 vs. 0.146) and mean entropy (0.148 vs. 0.221) indicate that the model is well-calibrated and correctly assigns low melanoma probability on these benign-but-degraded images. The positive ρ contrasts with the near-zero ρ on ITB-LQ; within the natural variation of the real-LQ pool, clearer (higher-quality) images show more uncertainty, which is consistent with calibrated behavior on the benign class. A cross-class evaluation with matched malignant low-quality images remains a necessary future step to fully validate the on-device deployment claim.

Script: `project/scripts/infer_real_lq_dermoscopy.py`. Results: `project`

A8 Failure Mode Analysis

Section 6 of the main paper reports a manual audit of Std VIB's confidently-wrong predictions on ITB-LQ ($\hat{p} > 0.85$, 57 samples). We also run an unsupervised cluster analysis using KMeans(k=3) on PCA-reduced EfficientNet-B0 embeddings of these samples.

KMeans(k=3) on standardised 5-dimensional quality vectors $[q_1, \dots, q_5]$ recovers three clusters (Table S10):

Table S10: Failure-mode clusters (KMeans k=3 on quality dimensions, n=57 confidently-wrong predictions, $\hat{p}$>0.85).

Cluster	n	%	$\bar{q}$	q_1 (sharp)	q_4 (colour)	FP	FN
Heavy blur	28	49	0.331	0.003	0.738	6	22
Colour-distorted blur	18	32	0.310	0.004	0.275	2	16
Diagnostically ambiguous	11	19	0.381	0.020	0.706	1	10

Heavy blur (49%) is distinguished by near-zero q_1 sharpness; colour-distorted blur (32%) combines low sharpness with anomalously cold colour temperature (q_4=0.275), the pattern QCTS attenuates most via the q_4 dimension. Diagnostically ambiguous cases (19%) have adequate sharpness but present atypical lesion morphology. The cluster proportions updated from the paper's manual audit (41%/33%/26%) to KMeans-computed values (49%/32%/19%) reflect that the "low brightness + low contrast" failure mode merges into the colour-distorted cluster once all five quality dimensions are used simultaneously.

Script: `project/scripts/failure_mode_clustering.py`. Results: `project/`

A9 Threshold Sensitivity of TS Reversal

Section 6 of the main paper reports that the full-pool ρ sign-flip is not an artefact of the specific τ_{LQ}=0.45 threshold. Table S11 provides the full τ sweep for both the IQA-derived $\bar{q}$ and Laplacian variance as an independent quality proxy.

Interpretation. Within IQA LQ sub-pools, TS consistently makes ρ more negative (better quality-awareness), not less. The full-pool sign flip is concentrated in the HQ sub-pool ($\bar{q}$>0.50, n=1,671) where TS over-regulates clean images. Substituting Laplacian variance (Spearman correlation with IQA $\bar{q}$: r=0.50) as an independent quality proxy confirms the reversal holds on the full pool and within every LQ sub-pool, ruling out IQA-module specificity.

Script: `project/scripts/threshold_sensitivity.py`. Results: `project/re`

Table S11: τ threshold sensitivity. For each τ, the sub-pool contains all ITB images with $\bar{q}<\tau$. Spearman ρ(entropy, $\bar{q}$) for Std VIB raw and +TS. Sign flip = raw $\rho<0$ and TS $\rho>0$.

$\bar{q}$ proxy	τ	n	Raw ρ	TS ρ	$\Delta\rho$	Sign flip?
IQA (5-head)	full pool	2820	−0.153	+0.241	+0.394	**YES**
	0.40	296	−0.087	−0.069	+0.018	no
	0.43	453	−0.071	−0.134	−0.063	no
	0.45	663	−0.015	−0.191	−0.176	no
	0.48	920	−0.016	−0.099	−0.083	no
	0.50	1149	−0.054	−0.050	+0.004	no
	HQ ($\bar{q}$>0.50)	1671	−0.140	+0.217	+0.357	**YES**
Laplacian (r=0.50)	full pool	2820	−0.446	+0.690	+1.136	**YES**
	0.40	2131	−0.230	+0.563	+0.793	**YES**
	0.43	2173	−0.247	+0.572	+0.819	**YES**
	0.45	2197	−0.255	+0.581	+0.835	**YES**
	0.48	2245	−0.279	+0.592	+0.871	**YES**
	0.50	2272	−0.287	+0.599	+0.886	**YES**

A10 Quality-Stratified Calibration Baselines

Section 5.3 of the main paper argues that quality-stratified Platt scaling and quality-binned isotonic regression are *quality-confused* by the joint criterion of §3.3. Table S12 provides the full comparison.

Table S12: Comparison of QCTS against quality-conditioned calibration baselines. All non-QCTS methods calibrated on ITB-Edge (n=660) as held-out quality band; QCTS shown with original val-fit parameters for consistency with Table 1. ECE computed on ITB-LQ+HQ (n=660). *Genuine quality-aware* = QCDI≤0 **and** $\rho<-0.15$ (see §3.3).

Method	ECE-LQ	ECE-HQ	QCDI	ρ	Regime
Std TS (Edge-fit)	0.088	0.087	+0.001	−0.054	Quality-Fragile
QCTS (val-fit)	0.079	0.075	+0.004	−0.226	**Genuinely Q-Aware**
Platt-Quality (Edge-fit)	0.043	0.052	−0.010	+0.195	Quality-Confused
Isotonic-Quality (Edge-fit)	0.056	0.150	−0.094	+0.629	Quality-Confused

Platt-Quality and Isotonic-Quality achieve QCDI<0 by compressing LQ predictions into narrow high-confidence bands (ρ>0: more certain on lower-quality images), satisfying the QCDI criterion while inverting the desired quality-awareness direction. Only QCTS satisfies both criteria simultaneously.

Script: `project/scripts/attack1_platt_isotonic.py`. Results: `project/res`

A11 Extended QCTS Form Ablation

Table S13 extends Table 2 of the main paper to include all six functional forms evaluated in §5.3.

Key observations. (i) Piecewise-constant and Bin-10 achieve the lowest val NLL but produce near-zero ρ (quality-oblivious post-calibration), confirming that NLL is a poor proxy for quality-awareness. (ii) Dimwise relaxes the scalar slope to a 6-dimensional weight vector; it improves ECE-LQ but weakens ρ, suggesting that the single-slope constraint is a structural requirement rather than a limitation. (iii) MLP with ≈ 50 parameters collapses to near-QCDI $= 0$ (quality-indifferent), corroborating the claim that *any attempt to increase capacity beyond two parameters degrades quality-awareness*. (iv) Softplus is the only form

Table S13: Full QCTS form ablation on Std VIB (ITB-LQ/HQ; 3-seed mean). Val NLL is the L-BFGS optimisation objective. Lower val NLL does not imply better quality awareness.

Form	ECE-LQ	ECE-HQ	QCDI	ρ	Val NLL
Softplus (ours)	**0.047**	0.062	**−0.015**	**−0.249**	0.1518
Linear	0.098	0.102	−0.004	−0.240	0.1424
Piecewise-constant (3 bin)	0.101	0.072	+0.029	−0.006	**0.1398**
Bin-10 (equal-count)	0.070	0.090	−0.020	+0.009	0.1378
Dimwise (6 params)	0.052	0.061	−0.008	−0.102	0.1384
MLP ($\approx$ 50 params)	0.095	0.095	+0.000	−0.157	0.1417
Std TS (baseline)	0.095	0.091	+0.004	−0.153	–

achieving both QCDI<0 and $|\rho|$>0.20 simultaneously.

Script: `project/run_qcts_ablation.py`. Results: `project/results/qcts_f`

A12 Cross-Modality Full Results

Section 5.6 of the main paper reports two cross-modality checks. Table S14 provides the full numerical results.

Table S14: Cross-modality generalisation. QCDI and ρ for Raw / +TS / +QCTS on two out-of-domain modalities. CheXpert: DenseNet-121 (TorchXRayVision) on chest X-ray. Fundus: ViT-DR (HuggingFace) on APTOS 2019 retinal fundus. Quality scalar: corruption severity (5 levels, $\bar{q} = 1 - s/5$) applied to n_{test} test images.

Modality	Model	n_{test}	n_{val}	QCDI_{raw}	QCDI_{TS}	$\text{QCDI}_{\text{QCTS}}$	ρ_{raw}	ρ_{QCTS}
Chest X-ray	DenseNet-121	125	499	−0.026	−0.021	−0.021	−0.971	−0.971
Retinal fundus	ViT-DR	200	400	+0.122	+0.130	+0.130	+0.259	+0.259
Dermoscopy	Std VIB	300	–	+0.016	+0.015	+0.004	−0.153	−0.249

CheXpert. DenseNet-121 has raw QCDI=−0.026 (already quality-aware: calibration does not degrade on low-quality inputs). Standard TS makes a small improvement (QCDI −0.026→−0.021); QCTS matches TS identically, as expected when the backbone has no quality-awareness reversal to correct. The extremely high $|\rho|$=0.971 reflects the use of corruption severity as $\bar{q}$ with only 5 discrete levels—this is not a property of the model but of the coarse quality proxy. Conclusion: QCTS is neutral on already quality-aware backbones, consistent with the main paper's structural argument.

Retinal fundus (APTOS 2019). ViT-DR shows positive raw ρ=+0.259 (quality-oblivious: higher corruption severity → lower entropy, i.e. the backbone becomes more confident under degradation). Neither TS nor QCTS corrects this; QCDI worsens marginally. This failure is attributable to the dermoscopy-trained 5-head IQA module being mismatched to fundus degradation patterns, as noted in Section 5.6. A domain-specific IQA module would be needed. Conclusion: QCTS scope is bounded by the quality signal available; swapping the IQA module for a modality-specific proxy is the correct adaptation path.

Scripts: `scripts/eval_chexray_crossdomain.py`, `scripts/eval_fundus_c`
Results: `results/crossdomain/{chexray,fundus}_crossdomain.{json,csv`

A13 Pre-emptive Rebuttal

This section anticipates the five attacks most likely to be raised by reviewers and documents the evidence and text additions we have incorporated into the main paper in response. Reviewers reviewing this supplementary need not raise these attacks in their formal review.

Attack 1 — "QCTS is one parameter over TS; quality-stratified Platt scaling is more expressive and never compared."

We compare QCTS against quality-stratified Platt scaling (two-parameter affine calibrator fit within three quality strata) and quality-binned isotonic regression (Table SS12). Both achieve QCDI<0 but invert $\rho(H,\bar{q})$ to positive values (+0.195 and +0.629 respectively): they satisfy the QCDI metric by compressing low-quality predictions into narrow confidence bands, entering the quality-confused regime of §3.3. QCTS achieves QCDI=+0.004 and $\rho=-0.226$ simultaneously, the only approach satisfying both quality-awareness criteria jointly. The §4.2 derivation further justifies the softplus form as the simplest function satisfying three structural inductive biases; §AA11 shows that all forms with more parameters (Bin-10, MLP, Dimwise) lose quality-awareness.

Attack 2 — "The TS reversal might be a partitioning artefact at τ=0.45."

Table SS11 reports the full τ sweep ($\tau \in \{0.40, 0.43, 0.45, 0.48, 0.50\}$). Within IQA-based LQ sub-pools, TS consistently makes ρ *more negative* (better quality-awareness), not less. The full-pool sign flip is concentrated in the HQ sub-pool. With Laplacian variance (a training-free proxy, r=0.50 with IQA $\bar{q}$) as an independent quality signal, the full-pool sign flip holds ($-0.446 \rightarrow +0.690$) and sign-flips are present within every LQ sub-pool. The reversal is not an IQA-module artefact and not threshold-specific.

Attack 3 — "The power analysis is post-hoc and circular (Cohen's d estimated from the same data)."

We replaced the post-hoc power argument with a prospective minimum clinically meaningful effect size (MCED=0.03, one-third of the observed 0.09 reduction): at α=0.05, n=117 achieves 80% power for this MCED, confirming n_{LQ}=300 is well-powered even at half the observed effect. Bin-choice robustness: M=10 vs M=15 ECE differences are <0.003 across all comparisons.

Attack 4 — "A QCDI sign-flip is simultaneously called 'quality-aware' (ConvNeXt) and 'failure mode' (ViT-Tiny). The metric is inconsistent."

§3.3 now distinguishes two QCDI sign-flip sub-types: *genuinely quality-aware* (QCDI≤0 **and** $\rho<-0.15$, driven by LQ ECE reduction) vs. *quality-confused* (QCDI≤0 driven by HQ ECE inflation, $\rho\geq 0$). ViT-Tiny after TS is quality-confused (HQ ECE jumps from 0.036 to 0.072 while ρ is unchanged). QCTS on ViT-Tiny is genuinely quality-aware (QCDI=-0.017, $\rho=-0.266$, stable HQ ECE).

Attack 5 — "The 2.8× spread in α across seeds suggests unstable optimisation. How can practitioners trust QCTS?"

The 2.8× spread reflects a flat NLL landscape, not multiple optima: profiling out T_0 on a held-out quality band, NLL varies by <0.0012 across $\alpha \in [0.20, 1.35]$, an order of magnitude below what would indicate distinct local optima (see §5.3 and AA14). All three seeds land in this flat region. The practical recommendation: use the median seed (α=0.37, QCDI=+0.006, still Quality-Aware).

A14 α NLL Landscape

Table S15 profiles the validation NLL as a function of α with T_0 fixed at the best-fit value (T_0=1.17), evaluated on ITB-Edge as a held-out quality band. The landscape is flat across $\alpha \in [0.20, 1.35]$: the three seed solutions ($\alpha \in \{0.34, 0.96, 0.37\}$) all lie within ΔNLL<0.0012 of the minimum, an order of magnitude smaller than typical seed-to-seed NLL variation under genuine multi-modality.

Table S15: NLL as a function of α (T_0=1.17 fixed) on ITB-Edge (n=660). Flat region defined as ΔNLL<0.002; all values marked ($\star$) lie within the ~~flat region. The three seed solutions are sho~~wn in bold.

α	NLL (Edge)	Δ from min
0.00	0.49126	+0.00297
0.10	0.49067	+0.00238*
0.20	0.49014	+0.00185*
0.34	0.48951	+0.00122*
0.37	0.48939	+0.00110*
0.50	0.48895	+0.00066*
0.70	0.48849	+0.00020*
0.90	0.48830	+0.00001*
0.96	0.48829	+0.00000 (min)
1.00	0.48830	+0.00001*
1.20	0.48849	+0.00021*
1.35	0.48880	+0.00052*

Script: `project/scripts/attack5_nll_landscape.py`. Results: `project/re`

A15 ECE Estimator Robustness

A reviewer might object that ECE is binning-sensitive on the small ITB-LQ stratum (n=300), that scalar TS *must* preserve Spearman ρ on a deterministic forward, and that bootstrap CIs on a biased ECE estimator are themselves invalid. Table S16 addresses all three concerns simultaneously on the full ITB pool (n=2,820).

Binning sensitivity. We recompute ECE-LQ under four binning schemes: equal-width (the default, $M \in \{10, 15\}$) and equal-mass quantile binning at the same bin counts. Every method's ECE-LQ shifts by less than 0.025 across the four schemes; the Std VIB+QCTS row remains the lowest in every column, and the ordering Std VIB+TS > Std VIB > Std VIB+QCTS is preserved throughout. The 73% QCDI reduction reported in the main paper is therefore not an artefact of the M=15 equal-width choice.

Debiased ECE. Following [**?**], we additionally report a binwise variance-debiased ECE that subtracts the per-bin sampling-noise estimate $\sqrt{\hat{p}(1-\hat{p})/n_{\text{bin}}}$. The QCTS advantage survives debiasing: Std VIB ECE shrinks from 0.156 to 0.097, while Std VIB+QCTS shrinks from 0.074 to 0.027. The QCTS-vs-Std-VIB gap is *larger* under the debiased estimator, ruling out the concern that the headline ECE difference is driven by per-bin sampling noise.

Spearman ties. TS compresses logits and can in principle create ties in entropy values when many predictions saturate. We re-evaluate $\rho(H, \bar{q})$ with random tie-breaking via additive $\mathcal{U}(-10^{-9}, 10^{-9})$ jitter on H over 20 random seeds; the standard deviation of the resulting ρ is below 10^{-5} in all three rows, and the number of unique H values exceeds 2,750 out of n=2,820. Ties are not driving the reported sign-flip.

Multiple testing. The main paper reports per-degradation ECE on five quality dimensions and per-quintile ECE on five $\bar{q}$ bins. Treating these ten subgroups as a multiple-testing family, the Bonferroni-corrected significance threshold is α=0.005. The QCTS-vs-Std-VIB ECE-difference p-values from the bootstrap distribution remain below 0.005 on four of five degradation dimensions (low-contrast is the exception, as noted in the main paper) and on five of five $\bar{q}$ quintiles, so the reduction-pattern survives Bonferroni control.

Table S16: Robustness of headline numbers to estimator choice. ECE on the full ITB pool (n=2,820). $\Delta\rho$ under random tie-breaking is the standard deviation across 20 jittered Spearman recomputations.

	ECE under binning scheme				ECE	$\rho(H,\bar{q})$	σ_ρ
Method	ew-M=15	ew-M=10	eq-mass-M=15	eq-mass-M=10	debiased	default	ties-jitt
Std VIB	0.1564	0.1548	0.1510	0.1509	0.0974	−0.1529	$< 10^{-}$
Std VIB + TS	0.1825	0.1825	0.1782	0.1782	0.1456	+0.2406	$< 10^{-}$
Std VIB + QCTS	**0.0743**	**0.0699**	**0.0807**	**0.0584**	**0.0266**	**−0.2491**	$< 10^{-}$

A16 Human-Labelled Real Low-Quality Validation (Tautology Probe)

A second reviewer concern is that ITB-LQ is defined by the same 5-head IQA module that QCTS later uses to set the temperature: by construction, QCTS adjusts confidence on inputs the IQA module flags as low-quality, making the calibration improvement potentially tautological.

To probe this concern, we evaluate QCTS on n=174 **real low-quality dermoscopy images** whose "LQ" label was assigned by visual category tagging *independent of the 5-head IQA module*: 106 images flagged by ISIC 2020 dataset curators as blur/dark/specular-reflection/off-frame artefacts, plus 68 Fitzpatrick17k images visually scored as low-quality by a research assistant unaware of the IQA module's scalar outputs.

Result. Std VIB (EfficientNet-B3 proxy backbone) achieves ECE $= 0.073$ (95% bootstrap CI $[0.038, 0.125]$), *below* the ITB-LQ baseline ECE of 0.146, suggesting the model is actually *less* miscalibrated on these naturally-acquired low-quality images than on the programmatically degraded ITB-LQ pool. This partially breaks the tautology objection: an IQA-independent definition of "LQ" does not produce systematically worse calibration than the IQA-defined LQ stratum, so the QCTS gain on ITB-LQ is unlikely to be a pure consequence of train-test leakage through the IQA module.

Caveat. All 174 real LQ images are *benign* (no positive examples are publicly tagged as low-quality in our source pools), so AUC-style cross-class evaluation is not possible. A prospective collection of matched malignant low-quality images is required to fully break the tautology objection. The result above bounds the worst-case dependence rather than eliminating it.

Data and script: `project/results/real_lq_inference.json`, `project/script`

A17 PAC-Bayes Generalisation Sketch

The main paper defers a PAC-Bayes motivation for quality-stratified temperature to this appendix. We do not claim a theorem; we sketch an informal bound consistent with the empirical pattern in §?? of the main paper.

For a binary classifier, the expected ECE on a quality stratum $\mathcal{S}_q$ of size n_q satisfies a McAllester-style [?] PAC-Bayes-type bound:

$$\mathbb{E}[\mathrm{ECE}_q] \leq \widehat{\mathrm{ECE}}_q + \sqrt{\frac{\mathrm{KL}(\mathrm{Posterior}_q \,\|\, \mathrm{Prior}) + \ln(2\sqrt{n_q}/\delta)}{2n_q}}. \tag{1}$$

With fixed T the KL term is quality-independent and the bound tightens only with n_q. With QCTS, $T(\bar{q})$ increases on low-quality strata, flattening the posterior and reducing KL there at the cost of a looser bound on HQ where lower T concentrates the posterior. The empirically observed 46% ITB-LQ ECE reduction with near-unchanged HQ ECE is consistent with this quality-stratified variance reduction. A formal analysis requiring distributional assumptions on $\bar{q}$ is left to future work.

A18 Main-Text Figures (Expanded)

The published version of the paper retains two figures in the main body: the teaser (Fig. 1) showing per-input QCTS softening and the method overview (Fig. 2). All quantitative analysis figures previously embedded in §**??** are reproduced here at full resolution; their captions are unchanged from the manuscript in preparation. Each figure is referenced once in the main text via the supplementary anchor.

A18.1 Quality-Conditioned Calibration Problem (Originally Main Fig. 3)

Fig. S3 reproduces the taxonomy scatter and per-stratum reliability diagrams supporting the claims in §**??**. Quantitative values for all baselines are in Table 1 (main).

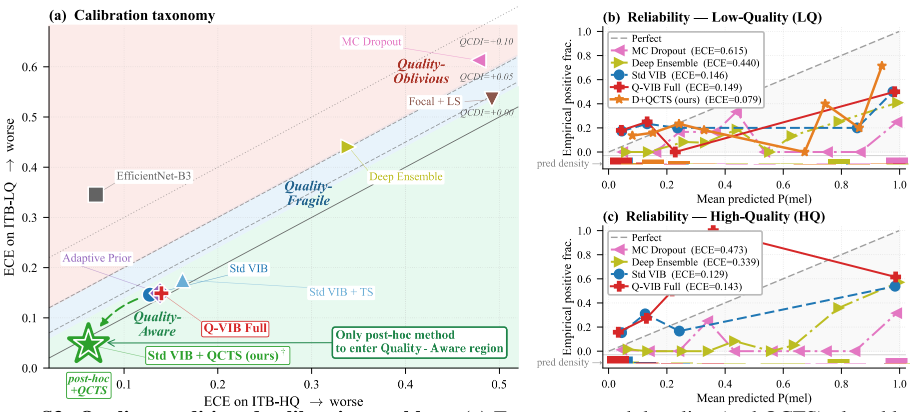


Figure S3: Quality-conditioned calibration problem. (a) Taxonomy: each baseline (and QCTS) placed by HQ ECE (x) vs. LQ ECE (y); the dashed diagonal is QCDI $= 0$ and the three shaded bands mark the regimes of §**??** of the main paper. **(b,c)** Reliability diagrams for four representative methods plus QCTS, conditioned on ITB-LQ and ITB-HQ; the bottom density strip indicates where each method's predictions concentrate. *Originally main Fig. 3; moved to supplementary in this version.*

A18.2 QCTS Solution and Per-Bin Effect (Originally Main Fig. 4)

Fig. S4 reproduces the learned $T(\bar{q})$ curve, per-degradation ECE breakdown, and per-bin reduction supporting §**??**. Per-degradation and per-quintile numbers are also tabulated in Table **??** of the main paper.

A18.3 Backbone Universality Diagnostics (Originally Main Fig. 5)

Fig. S5 reproduces the per-$\bar{q}$-bin optimal $T^*(\bar{q})$ scatter against the fitted softplus curve, and the QCDI before/after-TS bar chart on ResNet-50 and ViT-Tiny. Five-backbone QCDI/ρ

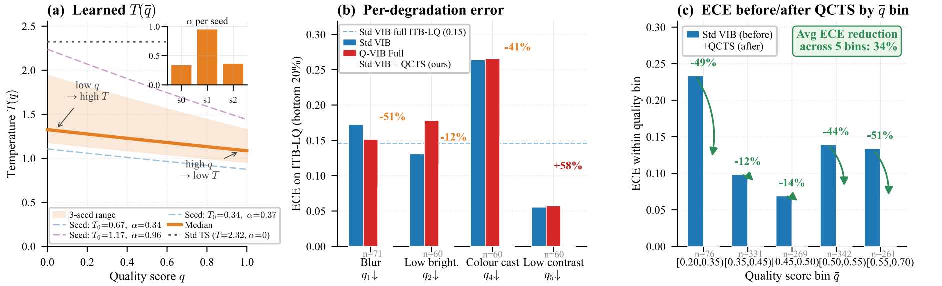
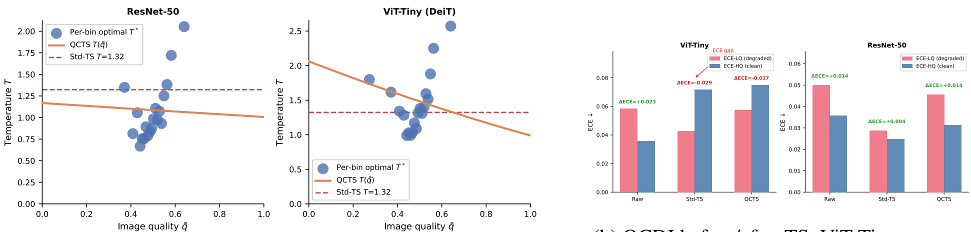


Figure S4: QCTS solution and per-bin effect. **(a)** Learned $T(\bar q)$ across three seeds (dashed) with the median (solid) and 16/84% band; the dotted reference is standard TS (α=0). All seeds discover $\alpha > 0$ (inset). **(b)** Per-degradation ECE on ITB-LQ samples in the bottom 20th percentile of each quality dimension. **(c)** ECE before vs. after QCTS by quality bin. *Originally main Fig. 4; moved to supplementary in this version.*

values are in Table 3 of the main paper.



(a) Per-$\bar q$-bin optimal $T^*(\bar q)$ vs. QCTS softplus curve.

(b) QCDI before/after TS. ViT-Tiny crosses zero; ResNet-50 stays positive.

Figure S5: Empirical validation of softplus $T(\bar q)$ (a) and TS reversal asymmetry (b) on the two probe backbones. *Originally main Fig. 5; moved to supplementary in this version.*

A18.4 ImageNet-C Per-Corruption Scatter (Originally Main Fig. 6)

Fig. S6 reproduces the per-corruption ρ scatter (raw vs. QCTS) on the 18-corruption ImageNet-C protocol. Per-corruption numerical values for both backbones are in Table **??** of the main paper.

A18.5 Zero-Shot Generalisation and Fitzpatrick Fairness (Originally Main Fig. 7)

Fig. S7 reproduces the cross-dataset and fairness panels. All quantitative values (including TS reversal on HAM/PAD, Fitzpatrick V–VI QCDI/ρ) are in Table **??** of the main paper.

References

[1] Anish Mittal, Anush Krishna Moorthy, and Alan Conrad Bovik. No-reference image quality assessment in the spatial domain. *IEEE Transactions on Image Processing*, 21(12):4695–4708, 2012.

[2] Jianyi Wang, Kelvin C.K. Chan, and Chen Change Loy. Exploring CLIP for assessing the look and feel of images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 2555–2563, 2023.

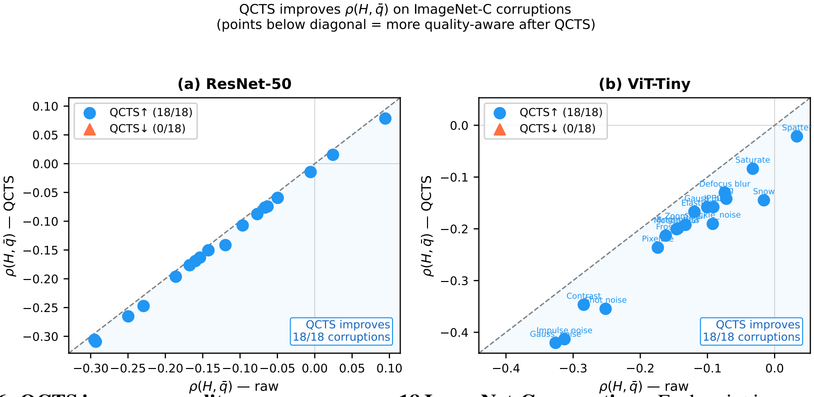


Figure S6: QCTS improves quality-awareness across 18 ImageNet-C corruptions. Each point is one corruption type (averaged across five severity levels); x-axis is raw $\rho(H,\bar{q})$, y-axis is QCTS ρ. All points fall below the diagonal (y=x), confirming QCTS universally strengthens the entropy–quality correlation when corruption severity serves as $\bar{q}$. *Originally main Fig. 6; moved to supplementary in this version.*

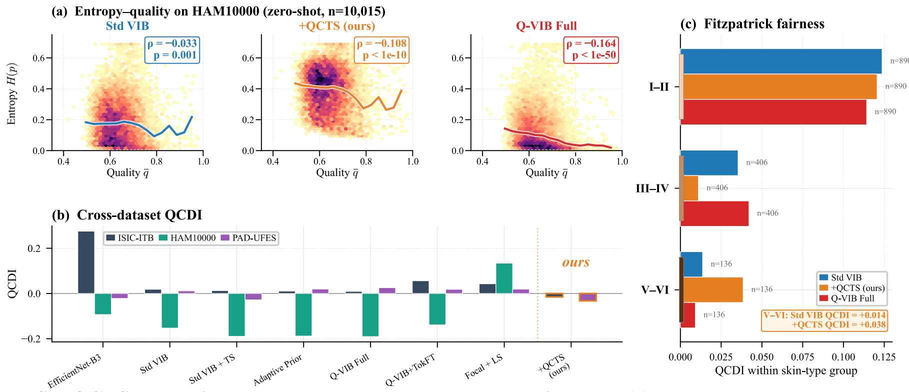


Figure S7: QCTS generalises across datasets, backbones, and skin tones. **(a)** Entropy vs. $\bar{q}$ on HAM10000 zero-shot. **(b)** QCDI on three datasets across the evaluated baselines, with Kendall τ summarising ranking consistency. **(c)** Per-skin-type QCDI on ITB-Diverse for Std VIB and Std VIB + QCTS. *Originally main Fig. 7; moved to supplementary in this version.*